

Alibaba B2B E-commerce Database Management System

Chapter 2

2.1 Introduction:

The entities required for the Alibaba B2B E-Commerce Database are identified based on the wholesale sourcing requirements outlined for the JBL Bluetooth earbuds case study. The attributes of each entity, along with their Primary Keys (PK) and Foreign Keys (FK), are defined to maintain proper data organization across Suppliers, Products, Customers, and Orders. The relationships between entities are analysed using cardinality and participation constraints, capturing how suppliers list products, how buyers place orders, and how order fulfilment, payment, and supplier feedback are tracked. This design forms the basis for the ER diagram, relational schema, and database implementation, and ensures data consistency, referential integrity, and reduced redundancy across the system.

2.2 Entity, Attributes and Keys:

i.. Entity: Supplier (Strong Entity)

Attribute	Key Type
Supplier_ID	Primary Key
Supplier_Name	Not Null
Location	Not Null
MOQ_Policy	-
Certification_Status	Check Key
Contact_No	Unique Key
Email	Unique Key

ii.. Entity: Category (Strong Entity)

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

iii.. Entity: Product (Strong Entity)

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Category_ID	Foreign Key
Supplier_ID	Foreign Key
Price	-
MOQ	Not Null
Listing_Type	Check Key

Key_Features	-
Units_Sold	-

iv.. Entity: Customer (Strong Entity)

Attribute	Key Type
Customer_ID	Primary Key
Business_Name	Not Null
Contact_No	Unique Key
Email	Unique Key
Country	-
Address	-

v.. Entity: Order (Strong Entity)

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Order_Status	Check Key
Total_Order_Value	-

vi.. Entity: Order_Item (Weak Entity)

Attribute	Key Type
Order_Item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Supplier_ID	Foreign Key
Quantity	Check Key
Unit_Price	-

vii.. Entity: Payment (Strong Entity)

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	Check Key
Payment_Date	-
Amount	-

viii.. Entity: Review (Strong Entity)

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Supplier_ID	Foreign Key
Rating	Check (1-5)
Review_Text	-
Review_Date	-

2.3 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Supplier	Lists	Product	One-to-Many	Partial / Total
Category	Categorizes	Product	One-to-Many	Total / Partial
Customer	Places	Order	One-to-Many	Partial / Total
Order	Contains	Order_Item	One-to-Many	Total / Total
Product	Included-in	Order_Item	One-to-Many	Partial / Total
Supplier	Fulfils	Order_Item	One-to-Many	Partial / Total
Order	Generates	Payment	One-to-One	Total / Total
Customer	Writes	Review	One-to-Many	Partial / Total
Supplier	Receives	Review	One-to-Many	Partial / Total